JAYPEE UNIVERSITY ANOOPSHAHR
Department of Mathematics

Mathematics for Data Science Lab

Lab Handout 9

Coin Toss Simulation and Graphical Analysis in Scilab

Programme: BCA **Software:** Scilab

Name: _____ **Roll No.:** _____

Learning Objectives

By the end of this practical, students should be able to:

1. Generate random numbers using Scilab.
2. Simulate repeated coin tosses.
3. Count Heads and Tails using logical conditions.
4. Calculate experimental probabilities.
5. Draw and interpret bar graphs.
6. Compare experimental and theoretical probabilities.
7. Plot the cumulative proportion of Heads.
8. Study the effect of increasing the number of trials.

1. Introduction

A fair coin has two possible outcomes: Heads and Tails. The theoretical probability of each outcome is

$$P(H) = P(T) = \frac{1}{2} = 0.5.$$

However, if a coin is tossed a finite number of times, the number of Heads and Tails may not be equal.

- **Theoretical probability:** The probability predicted by a mathematical model.
- **Experimental probability:** The probability estimated from observed outcomes.

The experimental probability of Heads is

$$P_{\text{exp}}(H) = \frac{\text{Number of Heads}}{\text{Total Tosses}}.$$

Similarly,

$$P_{\text{exp}}(T) = \frac{\text{Number of Tails}}{\text{Total Tosses}}.$$

In this practical, we will simulate coin tosses in Scilab and use graphs to analyse the results.

Representation

Throughout this practical:

$$\boxed{1 = \text{Heads}, \qquad 0 = \text{Tails}}$$

2. Important Scilab Commands

Command	Purpose
`clc`	Clears the console.
`clear`	Removes variables from the workspace.
`rand()`	Generates a random number between 0 and 1.
`rand(1,n)`	Generates a row vector of n random numbers.
`round(x)`	Rounds values to the nearest integer.
`sum(x)`	Calculates the sum of the elements.
`cumsum(x)`	Calculates cumulative sums.
`abs(x)`	Calculates absolute values.
`plot(x,y)`	Draws a line graph.
`bar(x)`	Draws a bar graph.
`xgrid()`	Displays a grid on the graph.
`xtitle()`	Adds a graph title and axis labels.

3. Practical 1: Generating Random Numbers

Aim: To generate random numbers and convert them into binary outcomes.

Program

```
clc;
clear;

x = rand(1,10);
disp(x, "Random numbers:");

tosses = round(x);
disp(tosses, "Simulated coin tosses:");
```

Explanation

1. `rand(1,10)` generates ten random numbers.
2. `round(x)` rounds each number to 0 or 1.

3. The resulting values represent simulated coin tosses.

Observation: The generated values are interpreted as Heads (1) and Tails (0).
Exercise: Change 10 to 20 and run the program again. Observe the output.

4. Practical 2: Simulating Coin Tosses and Calculating Probabilities

Aim: To simulate 100 coin tosses and calculate the experimental probabilities of Heads and Tails.

Program

```
clc;
clear;

n = 100;

tosses = round(rand(1,n));

heads = sum(tosses == 1);
tails = sum(tosses == 0);

p_heads = heads/n;
p_tails = tails/n;

disp(heads, "Number of Heads:");
disp(tails, "Number of Tails:");

disp(p_heads, "Experimental Probability of Heads:");
disp(p_tails, "Experimental Probability of Tails:");
```

Explanation

The expression `tosses == 1` identifies all outcomes that represent Heads. The `sum()` command counts the number of such outcomes.

The experimental probabilities are calculated as

$$P_{\text{exp}}(H) = \frac{\text{Heads}}{n}, \qquad P_{\text{exp}}(T) = \frac{\text{Tails}}{n}.$$

Verification

Verify the following two relationships:

$$\boxed{\text{Heads} + \text{Tails} = n}$$

$$\boxed{P_{\text{exp}}(H) + P_{\text{exp}}(T) = 1}$$

Record your observations below.

Quantity	Observed Value
Number of tosses	100
Number of Heads	
Number of Tails	
Experimental probability of Heads	
Experimental probability of Tails	

5. Practical 3: Bar Graph of Heads and Tails

Aim: To represent the frequencies of Heads and Tails graphically.

Program

```
clc;
clear;

n = 100;
tosses = round(rand(1,n));

heads = sum(tosses == 1);
tails = sum(tosses == 0);

counts = [heads, tails];

bar(counts);

a = gca();
a.x_ticks = tlist(["ticks","locations","labels"], ...
                  [1 2], ["Heads";"Tails"]);

xtitle("Frequency of Heads and Tails", ...
       "Coin Toss Outcome", "Frequency");

xgrid();
```

Interpretation

- The horizontal axis represents the outcomes.
- The vertical axis represents their frequencies.
- The height of each bar shows how many times the corresponding outcome occurred.

Questions

1. Which outcome occurred more frequently?

2. Are the frequencies necessarily equal?
3. What happens when the number of tosses is increased to 1,000?

6. Practical 4: Comparing Experimental and Theoretical Probabilities

Aim: To compare simulated probabilities with the theoretical probability of a fair coin.

Program

```
clc;
clear;

n = 1000;
tosses = round(rand(1,n));

p_heads = sum(tosses == 1)/n;
p_tails = sum(tosses == 0)/n;

experimental = [p_heads, p_tails];
theoretical = [0.5, 0.5];

comparison = [experimental; theoretical]';

bar(comparison);

a = gca();
a.x_ticks = tlist(["ticks","locations","labels"], ...
                  [1 2], ["Heads";"Tails"]);

xtitle("Experimental vs Theoretical Probability", ...
       "Outcome", "Probability");

legend(["Experimental";"Theoretical"]);
xgrid();
```

Interpretation

Each outcome has two bars:

- One represents the experimental probability.
- The other represents the theoretical probability, which is 0.5 for a fair coin.

The difference between the bars indicates how far the experimental result is from the theoretical value.

Questions

1. How close are the experimental probabilities to 0.5?
2. Why might the experimental and theoretical values differ?

3. Does the experimental probability always equal the theoretical probability?

7. Practical 5: Cumulative Probability of Heads

Aim: To study how the cumulative proportion of Heads changes as the number of tosses increases.

The cumulative proportion after k tosses is

$$P_k(H) = \frac{\text{Number of Heads in the First } k \text{ Tosses}}{k}.$$

For example, suppose the first five outcomes are

$$1,\ 0,\ 1,\ 1,\ 0.$$

There are three Heads. Therefore, the cumulative proportion after five tosses is

$$P_5(H) = \frac{3}{5} = 0.6.$$

Program

```
clc;
clear;

n = 1000;

tosses = round(rand(1,n));

cumulative_heads = cumsum(tosses);
number_of_tosses = 1:n;

cumulative_probability = ...
    cumulative_heads ./ number_of_tosses;

plot(number_of_tosses, cumulative_probability);
plot([1 n], [0.5 0.5], "r--");

xtitle("Cumulative Probability of Heads", ...
       "Number of Tosses", "Cumulative Proportion");

legend(["Observed Proportion"; ...
        "Theoretical Probability = 0.5"]);

xgrid();
```

Explanation of New Commands

- `cumsum(tosses)` calculates the cumulative sum of the outcomes.
- `1:n` creates the sequence $1, 2, \ldots, n$.
- `./` performs element-by-element division.

- `plot(x,y)` draws a line graph.

The second `plot()` command adds a horizontal reference line at the theoretical probability of 0.5.

Understanding the Graph

- The horizontal axis represents the number of tosses.
- The vertical axis represents the cumulative proportion of Heads.
- The horizontal reference line represents 0.5.
- The observed proportion may fluctuate above and below the reference line.
- With more tosses, the cumulative proportion generally becomes more stable, although it need not approach 0.5 steadily.

Exercise

Run the program for $n = 20$, $n = 100$, and $n = 1000$. Compare the resulting graphs.

Sample Size	Observations
20	
100	
1000	

8. Practical 6: Calculating the Absolute Error

Aim: To calculate the difference between the experimental and theoretical probabilities.
The absolute error is

$$E = |P_{\text{exp}}(H) - 0.5| .$$

Program

Add the following statements after calculating `p_heads` in Practical 2.

```
error_heads = abs(p_heads - 0.5);

disp(error_heads, "Absolute Error:");
```

Repeat the simulation with different sample sizes and record your results.

Number of Tosses	Heads	Experimental Probability	Absolute Error
50			
500			
5000			

Note: The absolute error does not necessarily decrease every time the number of tosses increases. Compare the overall behaviour rather than expecting every individual result to improve.

9. Independent Practice Exercises

1. Write a Scilab program to simulate 200 coin tosses and display the number of Heads and Tails.
2. Calculate the experimental probabilities of Heads and Tails for 500 tosses. Verify that their sum is 1.
3. Draw a bar graph comparing the frequencies of Heads and Tails for 1,000 tosses.
4. Draw a graph comparing experimental and theoretical probabilities for both outcomes.
5. Plot the cumulative proportion of Heads for 2,000 tosses and include a horizontal reference line at 0.5.
6. **Challenge:** Modify your program to calculate and display the absolute error in the experimental probability of Heads for different sample sizes.

10. Conclusion

The coin toss simulation demonstrates how probability can be investigated using computational methods.

The frequency bar graph shows the observed counts, the probability comparison graph shows the difference between experimental and theoretical values, and the cumulative probability graph illustrates how estimates change as more trials are performed.

A fair coin does not necessarily produce equal numbers of Heads and Tails in a finite simulation. Increasing the number of independent trials generally makes the experimental estimate more stable. In the long run, the cumulative proportion of Heads converges to the theoretical probability of 0.5.

Before Leaving the Lab

- ☐ Save your Scilab programs.
- ☐ Run all the practicals successfully.
- ☐ Save or capture the output graphs.
- ☐ Complete the observation tables.
- ☐ Answer the interpretation questions.
- ☐ Submit the program code and outputs in your practical file.

End of Lab Handout